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Sustainability transition in forestry transport: exploring business model challenges in a Swedish demonstration project

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ABSTRACT

Battery electric vehicles are recognized as a crucial pathway toward climate-neutral heavy transport. Yet their feasibility in forestry logistics remains underexplored, and existing research addresses it largely in technical and techno-economic terms. This paper presents initial findings from a system demonstration project aiming for 50% of new forestry trucks in Sweden to be electrified by 2030, engaging incumbent and emerging actors in co-developing a business ecosystem. Central to the long-term success of this transition is the evolution of viable business models for each participating actor. Drawing on business model innovation and transition management, we develop an integrative framework that locates each challenge both in an element of the business model – value proposition, value creation and delivery, or value capture – and in the level at which actors address it: operational, tactical, or strategic. Through 17 semi-structured interviews and a workshop with key stakeholders, we identify and analyze five categories of challenges: technical, economic, changes in roles and collaboration, logistics, and internal organizational. Our analysis reveals that challenges concentrate at the operational level, while the value proposition is only sparsely addressed. Furthermore, while the project has a collective vision, few organizations have a corresponding strategy. The paper contributes an analytical framework that distinguishes which parts of the business model come under pressure during a transition, and at what level. It also provides rare empirical data from the forestry transport context. It shows that collaborative project-level commitment does not by itself produce viable business models for the actors within it.

1. Introduction

Almost everything that moves goods by roads still runs on fossil fuels, a decade after 195 countries committed to reducing greenhouse emissions under the Paris Agreement (European Commission, 2015). The transport sector accounts for approximately 15% of global emissions (IPCC, 2023, p. 44), with the vast majority of its energy stemming from petroleum-based fuels such as diesel and gasoline (IEA, 2025, p. 128). Battery electric vehicles (BEVs) are widely regarded as a key enabler for climate neutrality in heavy-duty road transport (Shirizadeh et al., 2024), yet adoption has been slow, especially in demanding applications.

Forestry transport is one such application. In Sweden, domestic road transport within the forest sector accounts for 25% of all commercial road transport (Fossil Free Sweden, 2020), making forestry transport a significant contributor to national emissions. Forestry transport is also one of the most challenging to electrify, given the long haulage distances, heavy loads, unpaved forest roads, and dispersed pick-up locations, which strain vehicle range and charging infrastructure (Liimatainen et al., 2019; Lipiäinen et al., 2022). However, most studies focusing on barriers focus on technical and economic impacts, including vehicle range and battery weight (Lipiäinen et al., 2022), haulage distance and charging time (Liimatainen et al., 2019), vehicle performance and charging infrastructure (Larenas-Barra et al., 2022), and cost competitiveness (Engholm et al., 2024). Reviews report a similar emphasis, e.g. Nkesah (2023) finds that studies of road freight sustainability rarely examine collaboration between the actors involved.

This framing is incomplete, and the omission matters. Electrification does not enter an empty field, but an industry with a functioning system in place. The shift from diesel to electric unsettles existing arrangements and assumptions, putting pressure on how value is created, delivered and captured across a chain of interdependent actors – forest companies, transport purchasers, haulers, vehicle and charging developers, and drivers. Therefore, the question of how electrification scales in this sector is not only a technical one, but whether actors can construct business models in which electrified transport is viable for each of them (Darwish et al., 2025). Where these actors differ in resources, interests

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and institutional logics, the established power structures and value chains are challenged, placing new demands on both incumbent and emerging business models (Geels, 2012; Geels et al., 2017), requiring new business logic to be developed collaboratively and orchestrated across the network (Davis, 2016; Reypens et al., 2021).

This is a problem for multiple audiences at once. For firms, it determines whether capital committed to electrified fleets is recovered and how set goals of CO2 reduction can be met. For funders and organizers of demonstration projects – a common instrument of transition policy – it determines whether a sustainability-oriented pilot translates into industry-wide change. And for transport research, impacting how projected emission reductions materialize in due time. Yet, the role of the business model in transitions, and the mechanisms through which it shapes transitions, is not well understood (Bidmon and Knab, 2018).

This paper aims to identify the business model challenges facing actors in the early-stage electrification of heavy transport, and to account for how those challenges are distributed. Drawing on the business model's bridging role between technological niche and incumbent regime (Bidmon and Knab, 2018) and the distinction between operational, tactical, and strategic work (Kemp et al., 2007), we develop an integrative framework that analyses business model challenges in sociotechnical transitions. We apply this to a Swedish system demonstration project spanning seven experimental sites, which aims for 50% of new forestry trucks in Sweden to be electrified by 2030.

The paper makes several contributions. It develops the bridging role of the business model into an analytical device to distinguish which part of a business model is being reworked, and at what level. For transition management, it deepens understanding of how actors behave in the early transition phase. Empirically, it responds to the scarcity of research on BEVs in forestry transport. Doing so, learnings are valuable to transport and transition researchers, and to those designing, funding, or participating in demonstration projects.

The remainder of the paper is structured as follows. Section 2 reviews research on the electrification of heavy-duty transport and develops the integrative framework. Section 3 describes the case, data, and analytical approach. Section 4 presents the five identified categories of challenges. Section 5 analyses them through the framework, discusses their implications, and sets out the contribution of the study, before section 6 concludes.

2. Theoretical background

Understanding how an established industry electrifies its transport requires attention to both the concrete barriers that actors encounter and the deeper business model transformation those barriers signal. This section develops this understanding in two steps. First, we review what is known about the electrification of heavy-duty transport, and then specifically forestry transport. We then draw on business model innovation and transition management literature to build an integrative framework, in which the business model carries a new technology from niche experimentation toward a reconfigured regime.

2.1. Electrification of heavy-duty trucks

In the strive for sustainability, the decarbonization of transport is recognized as an important but particularly complex and challenging task (Biresselioglu et al., 2018; Salvucci et al., 2019) and has therefore spurred numerous interdisciplinary studies (Wandelt et al., 2025). In a systematic review of 75 articles, Nkesah (2023) found that measures using innovative technology, improved logistics and transportation processes, and policy-based measures can impact the overall economic, social, and environmental sustainability of road-based freight transportation, with the majority of articles investigating environmental and economic rather than social impact. The potential reduction of CO2 is clear. For example, Nicolaides et al. (2018) estimated that CO2 emissions from road freight transportation in the UK could be reduced by 91%-year 2040 through electrification, compared with 2016 levels. Such estimates should, however, be taken with precaution given the high levels of uncertainty in transport systems undergoing large transitions such as electrification (Nocera and Cavallaro, 2014). Furthermore, to their surprise, Nkesah (2023) found that prior studies on the electrification of road freight transport rarely involve collaboration among shippers. This is surprising given the important role of collaboration, cooperation, and competition in sustainability transitions. A lack of perspectives is problematic, given that the costs and benefits of electrification can vary significantly across stakeholders, as demonstrated by Kelly et al. (2025).

Research has identified barriers to the electrification of heavy-duty trucks, pointing out that they span a variety of areas, from technical, social, and economic areas, as well as the sociotechnical, socioeconomic, and techno-economic domains (Sugihara et al., 2023). Most barriers span multiple types, demonstrating the interconnections among technical, social, and economic aspects and indicating the need for a multiperspective approach to address them. For example, concerns about the reliability of the trucks to make it the full distance to the next charging station and about charging times address both technological and economic and social factors. Previous research has predominantly investigated technical aspects of the electrification of heavy transport (Sugihara et al., 2023). This includes, e.g. tradeoffs between payload capacity and battery capacity (e.g. Mareev et al., 2017; Ribberink et al., 2021), or how different charging scenarios impact cost competitiveness (e.g. Engholm et al., 2024). However, rather few studies have considered such factors in the context of business models. Other studies have focused on specific parts of the electrification system, such as for example the charging point operators and their role in creating the needed infrastructure, showing that emerging infrastructure system opens up new actors, roles and variety of business models for destination, terminal, semi-public and public charging stations (Gillström et al., 2024) or how logistics companies manage electrification of road freight transport (Jobrant et al., 2025). They also call for more research in this emerging area to understand the roles and how the systems are emerging. A parallel set of studies has examined business model aspects of other transport modes, such as the electrification of urban freight transport (Williamsson and Moen, 2022).

It seems truck manufacturers have only recently begun investing in BEVs for forestry applications. Besides the present project in Sweden, there are other examples from the UK (Williams, 2023) and emerging companies in Canada (Greenhalgh, 2025; Jarratt, 2021). Notably, however, sparse research has investigated the specific challenges of BEVs in forest transport. This scarcity is confirmed by the systematic literature review by Iyer et al. (2025), which identifies only three articles specifically examining the feasibility of BEVs for forest transport. Consequently, Iyer and colleagues included studies of other long-haul heavy-duty applications to draw conclusions for forest transportation. From a macro perspective, Lipiäinen et al. (2022) discuss visions for decarbonizing forestry transport in Sweden and Finland. While they find that transport is a significant source of CO2 emissions in both countries, they highlight the challenges posed by heavy batteries and the limited range of BEVs as significant limitations for forestry transport. These issues of a technical and logistical issues are echoed in Liimatainen et al. (2019), who compare the electrification potential of different transport modes in Switzerland and Finland, where the forest transport sector was particularly challenging in Finland, largely due to haulage distance. Overall, they find that on-road charging time and fast-charging technology significantly affect the potential of electric trucks. Larenas-Barra et al. (2022) interviewed 9 informants from transport companies, forestry companies, and truck distributors to investigate the potential of BEVs for log transportation in Chile. They found that current knowledge was limited, and the main challenges and risks included low vehicle performance, a lack of charging infrastructure, and insufficient incentives. Noteworthy, yet not further elaborated in the study, respondents envision economic sustainability as the major benefit (58%)

for BEV's in the forestry transport sector, compared with environmental sustainability (25%), not reported (i.e. missing background, 17%), or social sustainability (0%).

2.2. Organizing for business model innovation in industries undergoing a transition

The business model has become a central concept for understanding how organizations create value and how they must rework that logic when the conditions for creating value change. It provides an architecture of how organizations create, deliver, and capture value (Osterwalder and Pigneur, 2010; Teece, 2010) and can be seen as a guide or blueprint that aligns different strategic choices and operational activities to ensure economic viability. Over the past decades, a growing body of research has investigated how business model innovation functions as a key instrument for organizational transformation (Zott et al., 2011). For example, Teece (2010) argues that the more radical the innovation and the more complicated the *value capture*, the less likely traditional business models are to suffice. In the context of sustainability transitions, it is suggested that business model innovation provides a conceptual link among system elements, enabling more disruptive sustainability transformations (Schaltegger et al., 2016; Schmidt et al., 2026). While business models for sustainability share several characteristics of conventional business models, it is suggested that their activities need to be understood in a disaggregated form to comprehend their specific purposes (Lüdeke-Freund et al., 2024). Along these lines, in the present study, the business model value architecture specified in the literature matters not only as a static description of the firm's logic, but also as a way of distinguishing *which parts* of an actor's business logic come under pressure during a transition.

While much business model research has traditionally centered on individual firms, large-scale transition inherently unfolds across entire networks of interdependent actors. Such systemic change introduces increased complexity to business model innovation, as adjustments are not isolated but must be coordinated across multiple organizational boundaries (Adams et al., 2016). Transitions of this sort require firms to consider changes beyond their organizational boundaries, positioning themselves as one component within a larger system-wide shift to successfully meet the demands of the developing phenomena. Such transitions depend on collaborative knowledge creation across ecosystems, emphasizing exploration and knowledge sharing (Valkokari et al., 2022). Reflecting this, Cobben et al. (2022) call for integrating the multi-level perspective on niche-to-regime transitions with ecosystem and business-model research, an integration that this study employs to understand which parts of each actor's business logic are affected, as well as the dynamics among the actors involved in the transition.

Shifting transport in the forestry industry from diesel to electric is a large-scale transformation that requires changes at multiple levels, not only in technology but also in societal structures (Kemp et al., 2007; Loorbach, 2007). Existing frameworks, social cultures, and institutions must be reconsidered and reshaped, and such structural shifts need to be actively supported through coordinated innovation efforts. To facilitate such a transition, experiments and innovative projects play a vital role in co-evolution toward sustainable development. These projects help generate bottom-up movements while aligning with top-down policy visions and actions. Transition management highlights that actions at multiple levels in the socio-technical system must be coordinated and aligned (Kemp et al., 2007). Loorbach and Wijsman (2013) argue that the key concepts in transition management can also be applied at a firm level, they emphasize the need for joint reflection as well as indicators and measures to track the development in relationships toward the desired outcome of a societal transition. Whether transition efforts are led top-down by policymakers or initiated at the firm level, the fundamental pillars remain the same. Research in transition management also advocates for a multi-level perspective on socio-technical transitions, highlighting the various roles that projects have in the transition, going through phases such as experimentation, stabilization, diffusion, and finally reconfiguration (Geels and Locatelli, 2024). They call for more research into the role of projects in sustainable transitions and ask for projects that affect behavior and that understand the business case for the transition, motivating the present study.

According to Bidmon and Knab (2018), business models have an undisputed potential to systemic change, yet their exact role and the mechanisms by which they impact transition dynamics are not widely understood. In light of this, they highlight three different roles of business models. First, when embedded in the prevailing socio-technical regime, established business models can act as a *brake* on transitions, since they help lock in and stabilize the existing system. Second, positioned between the technological niche and wider regime, business models can also serve as a *bridge* – helping new technologies mature and eventually break through from niche experimentation into mainstream regime status. Third, viewed as a form of *non-technological niche innovation*, entirely new business models can themselves become drivers of transitions, forming a significant part of an emerging regime independent of any underlying technological breakthrough.

The literature reviewed above is usually kept apart. Business model research specifies the value architecture of the firm, but it is a static description that says little about where a firm sits within a system-level transition. Meanwhile, transition management specifies the levels and phases of change, but tends to treat the firm's value logic as a black box. We integrate the two by treating Bidmon and Knab's (2018) *bridging role* not as a description but as an analytical lens. In other words, if the business model is the intermediary that carries a niche-innovation into the regime, then the progress of a transition can be read from where actors are (and are not) doing business-model work. Furthermore, the work that they are doing can be seen from two dimensions, effectively intersecting the two literature streams.

The first dimension is the business model element under pressure, i.e. whether an actor is reworking its value proposition, value capture, or value creation and delivery (Osterwalder and Pigneur, 2010; Teece, 2010). The second dimension is maturation of a niche from experimentation toward regime reconfiguration (Geels and Locatelli, 2024), i.e. the timeline. Furthermore, drawing on transition management, the nature of business model work can be seen from three different types of activities (Kemp et al., 2007): operational innovation, the near-term experimentation and testing; tactical networking, where new collaborations and relationships are built; and strategic envisioning, where actors frame the system and the challenges ahead, see Fig. 1.

Taken together, the electrification of forestry transport is an early-stage, high-uncertainty transition in which the viability of both new and current business models is at stake. Yet, the specific business model challenges actors face, and where those challenges sit within their value logic, remains poorly understood. The framework developed offers a simplified way to analyze which parts of the business model are under pressure, how they are addressed during the transition from a diesel-based transport system to an electrified one, and what challenges are encountered in bridging the new technology from niche to regime. Applying this lens to a Swedish forestry-electrification pilot, we ask:

RQ: What are the key challenges in the early-stage electrification of forestry transport, and where in the business model are they addressed through operational, tactical, and strategic work?

3. Method

3.1. Research setting and design

The research setting of this project is a large national Swedish research project financed by Vinnova (TREE - TRansition to Efficient Electrified forestry transport, grant 2023-03185) focusing on the electrification of the forest industry. Swedish forestry transport hauls up to 74 tonnes, and the loads consist of either timber or wood chips. The national average haulage distance is roughly 100 km, but it is typically

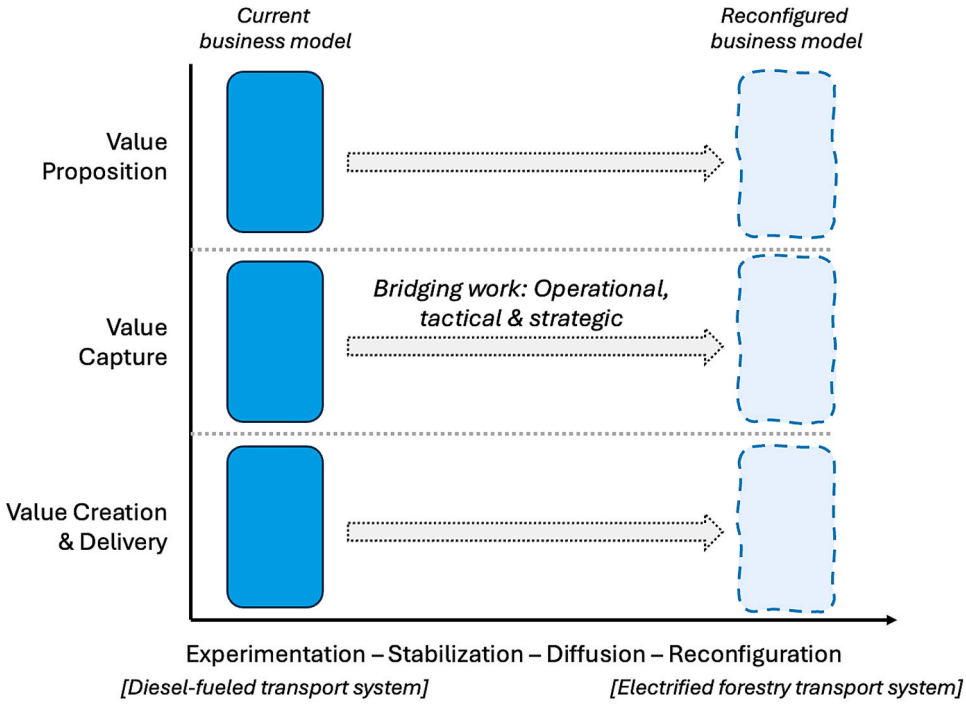


Fig. 1. The business model bridging a niche innovation across transition phases through operational, tactical, and strategic work.

longer in northern Sweden and shorter in the south. The project has 22 partners and involves all types of actors in the value chain, including the largest forest owners and organizations (transport purchasers), vehicle producers, haulers, transport coordinators, charging infrastructure companies, and route optimization software developers. In the project, twelve battery-powered electric trucks will be used in daily operations across seven sites, spanning from Småland in the south to Norrbotten in the north, which entails differences in weather and road conditions, route lengths, available charging infrastructure, ownership structures, contracts, and stakeholders involved.

Each site operates one or more BEVs for forestry haulage. Timber trucks, equipped with either a fixed or a detachable crane for self-loading at roadside landings, collect from harvesting sites that move as felling progresses. Chip trucks run more regular flows between sawmills, pulp mills, and terminals. Shift patterns vary across sites, with both single- and double-shift operations represented. The vehicles are provided by a single manufacturer predominantly in a 6x2 configuration coupled to five-axle trailers, totaling eight axles, with 624 kWh of installed battery capacity and charging at up to 375 kW. Charging arrangements differ across sites and combine private, semi-public, and public infrastructure. Most vehicles charge overnight at the hauler's depot or at a terminal, supplemented by opportunity charging during loading and unloading and by public fast charging along the route. Chargers in use or planned are rated between 40 kW and 400 kW.

The combination of a wide range of stakeholders and the exploration of a novel phenomenon (i.e., the electrification of the forest industry) motivates a qualitative approach (Cassell and Symon, 2004). Specifically, in-depth, qualitatively rich descriptions of individual perspectives and experiences are highly valued for uncovering the nuances of the developing phenomenon. Notably, the project spans four years, while this study captures the early stages, when several challenges remain to be solved and there are multiple uncertainties, rather than evaluating performance outcomes.

3.2. Data collection

The primary data for this study consists of 1) documentation from the first co-creation activity facilitated within the project, and 2) interviews with 17 informants representing different actors in the project. A reference group, well-informed about the purpose of the study and possessing good insight into the various types of actors within the project, was used to validate the interview guide (see Appendix A) and the purposive sampling of highly knowledgeable informants (Emmel, 2013), see Table 1 for an overview. When examining the table, the organizations' roles have been simplified for readability, e.g. "Transport purchaser" does not capture the full extent of their role in the network or business, and there are nuances between the different companies that are not captured by the shared labels.

The workshop involved 51 people, of whom 39 were from companies at one of the seven demonstration sites, and the remaining 12 were experts or researchers independent from the demonstration sites. Groups were formed, dividing the 39 participants among the seven demonstration sites they represented and mixing the different types of

Table 1
Overview of informants, their role, and the role of their organization in the network.

A	Business manager logistics	Transport company
B	Logistics manager	Transport purchaser
C	Purchaser, logistics and transport	Transport purchaser
D	Purchasing and transportation manager	Transport purchaser
E	Business developer industrial supply	Transport purchaser
F	CEO	Transport company
G	Sales manager e-mobility	Technology provider, vehicle producer
H	Managing partner	Technology provider, software
I	Co-founder, CEO	Transport company, charging point operator
J	Logistics manager	Transport company
K	Researcher forest operations business	Research and expertise
L	Researcher	Research and expertise
M	CEO	Best practice charging points
N	Project manager	Research and expertise
O	Researcher	Research and expertise
P	Transport service manager	Expertise forestry services
Q	Business developer	Expertise forestry services

organizations (e.g. transport purchaser, technology provider, etc.). The groups were asked to map out the actors, their relations, and material, data, transaction, or information flows between the different stakeholders, largely inspired by value network analysis (Allee, 2008; Galle and Matti, 2022), using prepared printed material. The 12 participants who were not assigned to a group were experts and rotated among the groups to spur the discussions. The purpose of this exercise was to establish a baseline understanding of the relations among the actors involved at each site, and to help identify key questions they need to discuss together, as well as the actors they need to involve throughout the project. All workshop material and notes were collected and digitized for documentation.

Following the workshop, follow-up interviews were conducted with selected participants. The aim of these was to obtain rich qualitative insights into the challenges and needs encountered in the early phases of developing their businesses in line with the goals of electrification in the forestry industry.

As a part of the interviews, the informant was asked to review a digitized version of the networks from the sites their organization is involved in, created during the workshop, and to reflect on them and the identified challenges in the creation and development of their business model(s). The interviews were conducted via video conferencing, lasted 60–90 min, and were recorded and transcribed verbatim.

3.3. Data analysis

The data analysis followed the thematic analysis described by Braun and Clarke (2006), combined with the coding methods of Miles et al. (2014). The interview transcriptions and workshop material were read through for familiarization. In a first coding cycle, over 200 initial codes (i.e. challenges) were identified using descriptive and in-vivo coding (Miles et al., 2014) to stay close to the informants' own formulations of the challenges they face. Codes were recorded in an Excel spreadsheet alongside the verbatim quote, informant identifier, and organizational role, which allowed challenges to be traced back to their source and further analyzed. No dedicated qualitative analysis software was used. The researchers coded one interview each, then compared findings and aligned on coding procedure through discussion, before the remaining interviews were coded by two of the researchers. In a second cycle, pattern coding was used to group related codes, and candidate themes were reviewed to ensure they accurately represented the included codes as well as the entire dataset (Braun and Clarke, 2006).

The process resulted in five themes of challenges related to business model development: technical, economic, changes in roles and collaboration, logistics, and internal organizational challenges. These were categorized into five overarching themes, see Fig. 2 below for the coding scheme. The analysis then moved abductively (Dubois and Gadde, 2002) between the inductively derived codes and the framework developed in

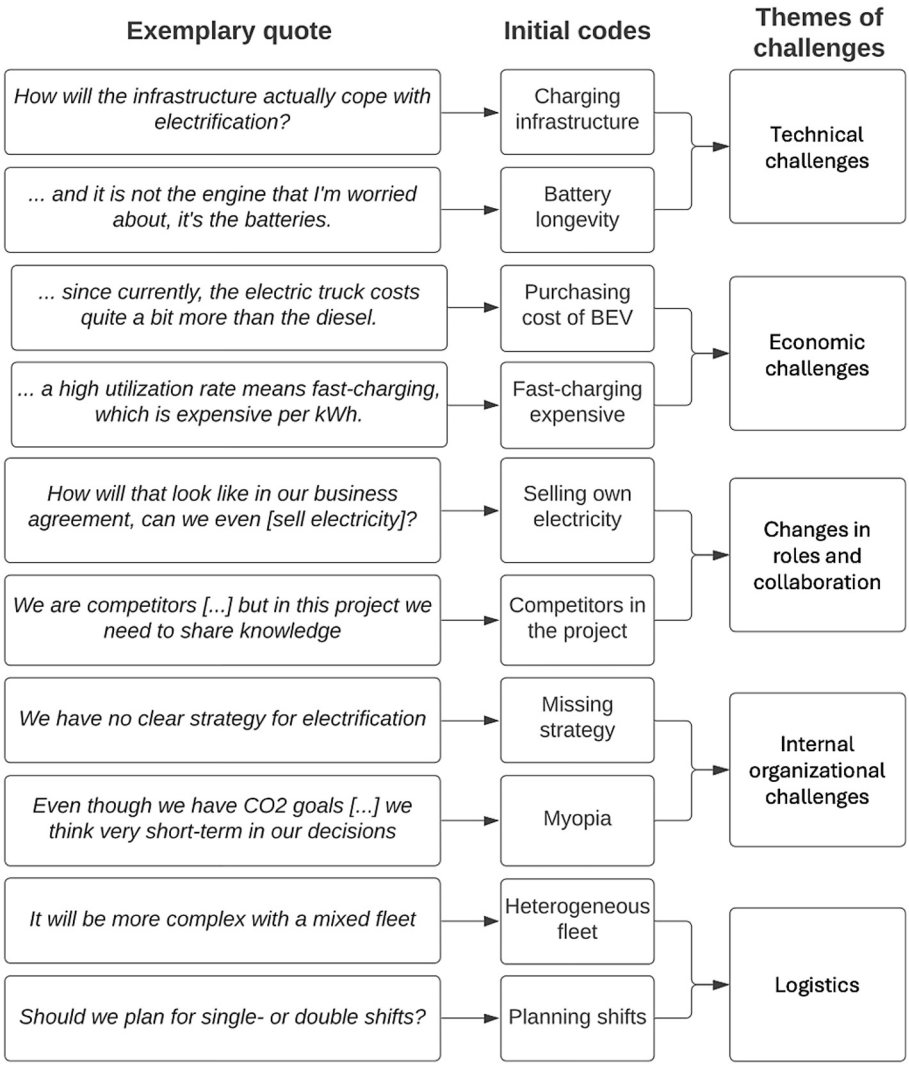


Fig. 2. Illustration of the coding process from quotes to initial codes and themes.

section 2, paying particular attention to elements of the business model (Teece, 2010) and operational, tactical, or strategic work (Kemp et al., 2007). Each theme was then positioned in a matrix of the business model elements and types of transition work, to visualize where challenges arise and at what level they are addressed (see Fig. 4).

For validation purposes, the preliminary results of the analysis were presented to informants for member checking (Miles et al., 2014, p. 270) and to all project participants, thereby inviting additional feedback and reflection.

4. Results

To give the reader a greater understanding of the different roles and to illustrate the different networks, a schematic illustration following the workshop is shown in Fig. 3.

As illustrated in Fig. 3, the workshop participants repeatedly identified uncertainty regarding the roles of charging actors, electric companies, and governmental bodies (i.e., decision-makers). Notably, one of the companies jointly owned by industry members (both forest owners and purchasers) was found in only one of seven site mappings. This is surprising, as they hold a position as an impartial party providing support for the majority of forestry transports in Sweden. Furthermore, the driver is missing as a distinct stakeholder.

In the following sections, the identified challenges are described and categorized into the themes of technical, economic, changes in roles and collaboration, logistics, and internal organizational challenges. Then, section 5 rests on how the challenges within the themes are distributed across the elements of the business model and the levels at which actors address them.

4.1. Technical

Technical challenges concerned the vehicle's capability under forestry operating conditions, the availability and reliability of charging infrastructure, and the capacity of the electricity grid at locations where charging is needed.

We will load them fairly heavily. Nobody has done long-term testing of these trucks, they haven't existed. Here, we will be driving in shifts, heavy loads, out in the forests, with quick charging all of the time… How does that affect the vehicle and battery capacity? What is the secondhand use of these early trucks?

– Informant F, CEO, Transport coordinator.

Concerns about the vehicle mostly center on the battery's unknown performance in this relatively unique setting. For example, the roads on which several routes are planned are often unpaved, and the trucks will operate in cold weather for much of the year (especially at the most northern sites). Consequently, the loading capacity of the vehicle is unknown and may vary significantly depending on contextual factors relating to the route (e.g. road properties) and the weather (e.g. temperature, wind, ice/snow/mud). Given these parameters and the newness of the technology, the technological lifespan of the (expensive) battery is therefore uncertain. Some also highlight a potential risk in different technologies and standards being developed, e.g. for data sharing or charging, which could lead to complications for early adopters when another dominant design emerges.

Similarly, concerns about charging relate to power production and grid capacity, technical issues with chargers, and logistics of the charging infrastructure. While the charging infrastructure for heavy transport is expanding along main roads in Sweden, forestry transport routes are typically far away from these. There are also concerns that a large-scale transformation to electrified trucks may be constrained by power grid capacity. Besides the longer time required to charge the trucks (compared with filling up with diesel), one informant with experience owning and managing an electrified truck described issues with the charger. Specifically, when the charger was left without surveillance, it occasionally stopped charging, which was believed to be due to temporary power surges. This technical challenge led to significant consequential challenges for the following work shift.

4.2. Economic

Economic challenges include the acquisition costs and financing of the vehicles, uncertainty about residual values and the secondary markets for trucks and batteries, the volatility of electricity prices, and the

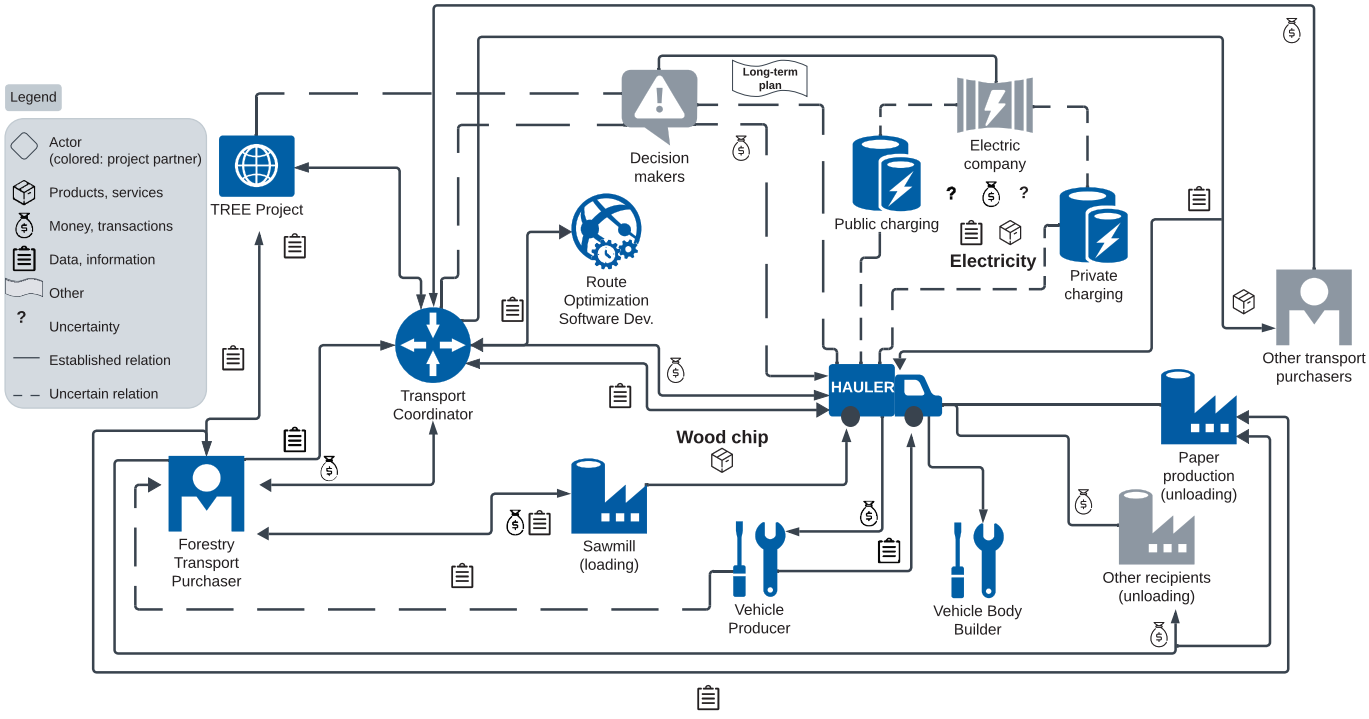


Fig. 3. Schematic illustration representing one of the sites.

distribution of resulting additional costs and risks between transport purchasers and haulers.

And in an industry with, generally speaking, very, or… Yes, I'd say very low profitability as the transport industry has, the hauler does not want to take that cost either, because the calculations simply don't make sense.

– Informant F, CEO, Transport coordinator.

To date, the new electrified trucks are significantly more expensive than their combustion-engine counterparts. From the technology developer's perspective, current low production volumes and fluctuating material prices have a significant impact on the listing prices of the trucks. As part of the project, special arrangements are made to partially cover these investments, but informants expressed concerns about how prices may develop over time and whether haulers will be able to justify the investments, given that many of them operate on low margins. This was particularly concerning since there is a large complexity in accurately creating an investment basis for the trucks, as there are currently unknown factors. This includes, for example, the residual value of the truck and the battery, the possible utilization rate, and whether there are other applications for the battery after it is no longer fit for use in the truck.

The electricity price is harder to manage contractually than the diesel price. Today, there are often agreements that allow for changes in diesel prices, which are adjusted monthly. In contrast, the current price of electricity fluctuates significantly over a much shorter time span and varies across the country. Consequently, new arrangements are required to cover the costs and risks involved between the transportation companies and transport purchasers (i.e. forestry companies). Regarding charging, some sites will rely solely on fast-charging stations, which are open to the public, thereby avoiding investment and instead paying a higher price per kWh.

Informants expressed concern about where the additional cost of electrified transport would ultimately settle. Haulers operate with thin margins and cannot carry it alone, while transport purchasers compete against firms still running on diesel. Consequently, if the large-scale transformation is to succeed, the higher costs of electrified transportation will need to be reflected in the retail pricing of goods for end customers who may be reluctant, as highlighted by informants.

4.3. Changes in roles and collaboration

Challenges relating to roles and collaboration concerned the ownership and sharing of data and the standards governing it, contractual and ownership arrangements including leasing and electricity clauses, imbalances in resources and influence between actors, and uncertainty about which roles and actors the electrified system will require.

Our haulers are in an alliance with us and put the future of their business in our hands. If we do bad business, well, then their company will do bad business.

– Informant A, Business manager logistics, Transport coordinator.

Transport of forestry goods typically relies on numerous haulers, many of which are relatively small entrepreneurial companies. This industry is described by informants as having low margins and slim organizations, which in turn means that few resources can be spent on development. Therefore, there is a risk of unintentionally creating imbalances in which larger haulers have more power and opportunities to be actively involved in electrification and its implications for new business models.

There are also concerns about potential resistance from drivers, as raised during the interviews, despite their absence from the mapping in Fig. 3. Specifically, the range of the electrified truck will depend more on the driver than that of a diesel truck, due to the longer charging time. For example, collecting timber with a crane and avoiding getting stuck on narrow, icy forest roads places requirements on the driver's competence. Without experience, a driver may get stuck or need more time, leading to a time-consuming stop to charge earlier. The drivers will also make the final decision on when and how to charge the truck, where a less experienced driver may, for example, choose to charge when 30% of the battery remains, whereas a more experienced driver could choose to complete the drive with the same charge. At the same time, there is a shortage of experienced drivers, leaving them with a lot of say in truck selection. This may be problematic, as informants describe a culture within the industry in which drivers highly value powerful and proven diesel engines. In addition, the drivers' work environment will change significantly, including changes to work hours and new types of decisions to make.

Notably, there are also competitors in the TREE-project, which was described as a new way of working. Given that the project is still in the early phase, there have been no lock-ins yet, despite clear conflicts of interest and a need for data sharing between partners. This was seen as largely enabled by a joint ambition to contribute to more sustainable forestry transport and a clear division between project-related work and daily operations.

Given that access to cheap electricity significantly impacts the costs of operating the electric trucks, there are discussions and ideas concerning how large transportation purchasers could provide this to haulers through producing electricity at key locations (e.g. wind-power, solar energy, battery storage). However, there is a large complexity in such offers since the haulers may also take on jobs for competing firms.

Hauler business models are described as well-established and as using similar pricing strategies throughout the industry. In terms of creating new business, informants see the electric truck as taking on the same jobs as the existing fleet, yet with a need for more long-term relationships and contracts as a way to mitigate certain risks. Most of these relate to fluctuating prices in electricity, and the residual value of the truck or its battery, outlining ownership.

4.4. Internal organizational

Internal organizational challenges, which were fewer than in the other categories, concerned the absence of internal strategies and subgoals, organizational structures and processes built around existing operations, and questions of competence and attitude, particularly among drivers.

There are expectations that a lot of things should happen soon. And there, I feel that there's a need to carve out some type of clarity. We have no clear strategy in how we think about this, instead we hope to learn a lot from this project….

– Informant B, Logistics manager, Transport purchaser.

The project has a set goal where 50% of new forestry transport vehicles should be electrified by 2030. However, there is a lack of internal strategies or subgoals related to these, which one interviewee attributes to uncertainty at this point in time regarding technical development and infrastructure, as well as to myopia among politicians. This lack of a clear strategy and goals was believed to slow down collaborative efforts. Conversely, clearer internal goals could be a remedy in the face of uncertainty and difficult investment basis.

Interviewees described organizations whose routines, systems, and competencies are configured around diesel operation. For haulers, there are clear organizational challenges related to the need for extended working hours and access to more drivers with the right competence, when there is currently a scarcity of drivers. For technology developers, existing sales channels have a fixed structure with several links between the manufacturer and the end user, which could hinder business model innovation if not all parts are included.

4.5. Logistics

Logistics challenges included planning routes under new parameters, organizing shifts and working time, and scheduling charging, including access to and queuing at chargers.

Today, you can fuel a timber truck in ten minutes with 300 L of diesel.

Now, you don't know if there is a queue to get access to a charger, nor how much charge you can get.

– Informant C, Purchaser – transport and logistics, Transport purchaser.

Presently, there is a highly sophisticated system accepted by the forestry transport industry that selects acceptable routes based on a set of parameters (e.g., CO2 emissions, driving time, etc.), and the driver must not deviate from it if the hauler is to be paid. However, with electrified trucks, there are new parameters to consider. This could include, for example, the availability of chargers at a given time, the type of chargers available, or the charging from regenerative braking during long downhills on specific road choices.

The routes chosen for experimentation differ across sites, as some actors have a clear set of frequent routes while others have highly variable ones. The former could be wood chips from sawing mills transported to pulp mills, whereas the latter is more common for picking up timber in the forest. Consequently, some have selected specific routes to be operated by a BEV truck, and others are more openly investigating the potential of the BEV truck and which transportation it can replace. After dropping off a load, trucks are often tasked with picking up a return load near the drop-off location. However, as feared by informants, a BEV truck that has been driving for hours may not have sufficient charge to pick up such loads, significantly reducing the time spent with a full load.

Regulated driving hours were written around an operation in which refueling is brief. Charging takes substantially longer and must be fitted within the same regulated shift, so the time previously available for driving is now consumed by charging. Many of the trucks will be driven across multiple shifts to increase the utilization rate, which is new for several actors, and contingent on the opening hours of drop-off locations.

5. Discussion

Alternative fuels are transforming the transportation industry, yet few studies have examined their feasibility in the forestry sector, their effects on business models, and the strategic implications for existing organizations. This study aims to provide a comprehensive and qualitatively rich view of how the forestry transportation industry is navigating the shift toward electrification, thereby contributing to understanding the challenges for the adoption of BEVs and the complexities of sustainability transitions in industries.

The electrification of heavy road transport is a societal transition: vehicles, energy infrastructure, logistics, and the organizations that operate them must be reconfigured together, and the intended emission reduction materializes only when they are. Within that broader shift, this study identifies five interrelated categories of challenges that organizations face during the early stages of electrifying the forestry transportation industry: technical, economic, changes in roles and collaboration, logistics, and internal organizational challenges. These are interrelated in the sense that, for example, a technical concern about range is simultaneously a logistics concern about which routes are feasible and how shifts can be organized, because charging duration enters the schedule in a way refueling did not. Similarly, questions about roles and collaboration can be inseparable from economic ones, e.g. deciding who charges, who owns the infrastructure, and who bears the cost of price volatility is the same decision.

Previous studies have investigated how electrification of heavy transport impacts specific elements of business models, e.g. logistics or cost competitiveness (Engholm et al., 2024; Jobrant et al., 2025) or reviewed existing literature (Iyer et al., 2025). Our study corroborates these with rich qualitative insights in the context of forestry transport and sheds light on the implications and challenges for business model development in the electrification transition.

Revisiting the framework developed in Section 2 (see Fig. 1), we now analyze the challenges described by informants based on the elements of the business model and in the type of work through which actors address them, unfolding the bridging role of the business model (Bidmon and Knab, 2018) and Kemp et al. (2007). Crossing the two results in the matrix in Fig. 4, where each balloon represents a theme of challenges. Because challenges within a theme span more than one cell, the position shows the dominant pattern rather than the full spread.

To illustrate how the framework separates out what a single empirical concern entails, we consider two themes in more detail: technical and logistical.

First, technical issues typically occur in the operational space, concerning value creation and delivery. For example, informants described being geographically tied to the few chargers that exist, with uncertainty about available power and range under load. In industry terms, haulers collect timber at landings in various locations, without a grid connection. Theoretically, this is a resource base for creating the service, currently addressed operationally because it is encountered daily as a constraint with current landing sites, rather than what can be achieved in the long term.

Second, logistical issues are typically handled operationally and relate closely to value creation and delivery. For example, refueling a diesel timber truck takes around 10 min, while recharging a BEV takes substantially longer. This issue must be fitted within regulated driving hours and the opening hours of reception points. Industry-specifically, high vehicle utilization rates are required by transport purchasers, further compressing the time window. Theoretically, this is value delivery, i.e., the same service reorganized around a new constraint, and resolved within the hauler's own planning.

Taken together, the challenges are interrelated and not isolated to individual firms, interdependencies between vehicle technology, charging infrastructure, haulers, forestry companies and drivers introduce additional complexity and inertia. As highlighted in the examples above, charging is a challenge that appears at all levels in our material, though it takes a different form at each. Operationally, it is a question of using the chargers that currently exist, i.e., where they are, how long charging takes, and how much working time is consumed waiting. Tactically, it becomes a question of providing charging, i.e., whether a depot charger can be justified for a single vehicle, who owns it, and how access is shared and booked where several operators depend on the same charger. Strategically, it is a question of whether a charging network and sufficient grid capacity will exist at all, which informants describe as a development that is uncertain. The same constraint, therefore, requires three kinds of work by progressively wider constellations of actors. However, in our material, the attention the constraint receives declines as the constellation widens.

Bidmon and Knab (2018) position the business model as the intermediary through which a niche innovation is carried toward the incumbent regime, but treat the business model as a single whole. Our analysis reveals that roughly half of the individual coded challenges concern operational work, whereas the value proposition is only sparsely addressed. In other words, read through the bridging lens (Bidmon and Knab, 2018), the informants predominantly focus on the operational and, to a lesser extent, tactical rework of value creation, capture, and delivery. In contrast, as the thinly populated value proposition row indicates, the actors are doing the execution work of running electrified transport while paying significant attention to the strategic rework of what they offer and to whom they offer it.

This is surprising, as the value proposition is suggested to be the element that Teece (2010) argues to be foundational, a close link to business strategy, and one that anchors the other elements. We are, however, cautious about attributing this to mere oversight. First, operational dominance may be more appropriate in the current phase of the transition in order to run the pilot, whereas strategic reconfiguration is a characteristically later activity (Geels and Locatelli, 2024). Second, the incumbent business models constrain what can be considered. Bidmon and Knab (2018) suggest they stabilize the existing system, and Chesbrough and Rosenbloom (2002) describe them as a cognitive filter, screening out alternatives that do not fit established logics. Our material shows this directly, BEVs is understood as taking the same job as the

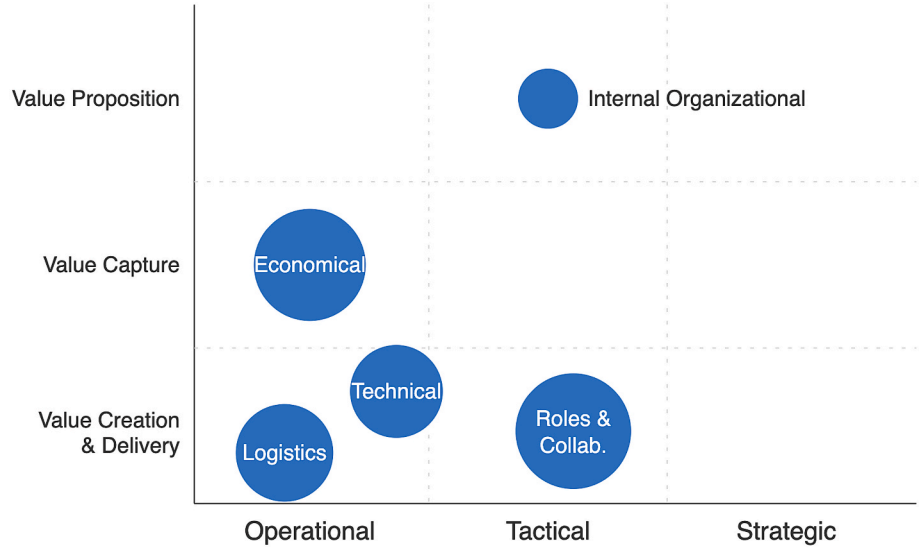


Fig. 4. The five identified challenge themes in relation to business model elements (Teece, 2010) and transition management (Kemp et al., 2007).

existing fleet, with new arrangements sought to mitigate risk rather than new business opportunities. For these reasons, our informants may be more inclined to reason about the tensions in e.g. value capture logic that stabilizes their current business model rather than radically innovating the more foundational value proposition (Osterwalder and Pigneur, 2010; Teece, 2010). Third, industry specifics reinforce this. Forestry haulage is described by informants as a well-established industry, using similar pricing strategies across the sector and with haulers operating on thin margins, thereby leaving individual actors little room to introduce novelty. Furthermore, haulers typically take on work for competing purchasers, which complicates new arrangements that would not be uniformly beneficial, e.g. a transport purchaser selling electricity to a hauler that also carries timber for a competitor. Finally, a methodological contribution cannot be excluded. For example, because operational difficulties are more concrete and easier to articulate than diffuse strategic questions, the latter require greater seniority among informants.

The uneven distribution is not explained at the level of the individual actor. The following two sections, therefore, look wider. The first section highlights the relationship between the project's collective ambition and the strategies of the actors within it, and then the dynamics that shape where actors direct their attention.

5.1. Scaling the transition: Collective ambition under institutional constraints, without actor-level strategy

The present project develops new knowledge around emerging phenomena through collaborative exploration and knowledge sharing (Cobben et al., 2022; Valkokari et al., 2022). At the project level, there is a clear goal orientation (Geels and Schot, 2007) with a vision and goal of 50% of new forestry trucks in Sweden being electric by 2030. However, we find that this does not necessarily translate to individual actors. Instead, there is a lack of internal organizational goals, strategies, or roadmaps concerning how they relate to electrification. This dissonance between collective ambition and individual actors' preparedness suggests that systemic alignment is still in its early stages and reflects the high levels of uncertainty this transition entails.

Transitions in socio-technical systems are rarely driven by technology alone—they require concurrent shifts in governance, organizational structures, and institutional support (Loorbach, 2007). Our results show how unevenly that support is currently perceived and whether public decision-makers have a clear idea of how to facilitate the transition away from fossil-based fuels, as highlighted in Fig. 3. Thus, the alignment between the project and the wider policy level is unclear (Kemp et al., 2007), making investments in BEVs risky for these early adopters. Policy is widely held to support transitions by altering economic conditions through subsidies, regulation, taxation, or standards (Geels and Schot, 2007). Our findings are consistent with this, but they also point to constraints that no subsidy addresses. Regulated driving hours are a case in point: the rules were written around diesel operation, where refueling is brief, and they become a binding constraint when charging takes significantly longer. In other words, an arrangement built around the incumbent regime that continues to govern the niche. Therefore, the institutional support for electrified forestry transport needs to extend beyond financial instruments to the rules that structure how the work is done.

5.2. Dynamics of the transition to electrified forestry transport: Time horizons, power asymmetries, and the role of drivers

Near-term cost challenges were articulated by informants in fine detail, e.g. investment in the electric truck or fluctuations in electricity price, as also investigated by e.g. Engholm et al. (2024). In contrast, far less attention was given to new business opportunities that an electrified fleet might create, and the strategic implications for their own organization. This contrast is telling, costs can be calculated against the present technology and business models, whereas new value propositions through electrification are radically different. The present model, therefore, defines the terms by which the transition is reasoned about (Chesbrough and Rosenbloom, 2002), and those are cost-facing and quantitative in this case. How different actors make sense of the transition (Gioia and Chittipeddi, 1991; Weick et al., 2005) is an interesting avenue for further research.

The feasibility and legitimacy of electrified transport could fall short if there are ambiguities in roles and a lack of governance (Darwish et al., 2025). The power relations among different actors are frequently discussed in the literature in relation to various forms of ecosystems and transitions (e.g. Cobben et al., 2022; Geels, 2012). The tactical work in our material is precisely where such asymmetry will come into play, e.g. new contractual arrangements, or the distribution of cost and risk between transport purchasers and haulers. In the demonstration phase, these questions remain largely open because investment risk is shared or subsidized, and volumes are small. Whether these asymmetries become consequential depends on how costs and risks are distributed once pilot-phase support ends, and volumes grow. Future studies into the role of power relations among different actors (see e.g. Kaplan, 2008) are

suggested to better understand how a transition toward electrified forestry transport unfolds amid high uncertainty.

The role of truck drivers is a further aspect that our interviews bring out as overlooked. As stated, the haulers are relatively limited in resources and capabilities. Yet, the drivers have a lot of influence and are impacted by the transition (e.g. recharging time, less noise, reduced vibrations, etc.). There is currently a shortage of truck drivers in Sweden (Iyer et al., 2025), and forestry transportation requires highly skilled drivers due to the unique operating conditions, according to informants. This should be considered in relation to the survey by Larenas-Barra et al. (2022), in which none of the respondents reported beneficial social sustainability implications for BEVs in forestry transportation, even though the adoption of electric automobiles has a significant impact on users (Gonçalves et al., 2026). Consequently, we suggest future research should include the role of drivers in e.g. purchasing decisions, studies on the acceptance of BEVs for the users in forestry transport, as well as the overall market interest and sales strategies for electrified forestry transport.

Taken together, the challenges we identify are not those of a technology that does not work. Instead, they are the challenges of fitting emerging technology into an industry that already has a functioning system in place, where the costs, contracts, routes, and shift patterns are built on the assumption of diesel fuel. Our case shows that electrification clearly unsettles the arrangements through which value is created, delivered, and captured across a chain of interdependent actors. Our informants are working on resolving those disturbances, but what remains largely unaddressed is the prior question of what electrified forestry transport is for: what it offers, for whom, and on what terms. Our framework places that question at the strategic end of the value proposition, and our evidence suggests that demonstration projects should explicitly make space for it, alongside the technical and operational testing they are designed to conduct.

5.3. Theoretical and practical contributions

The study makes four distinct contributions with implications for practice. For the business model innovation literature, the study qualifies the bridging role that Bidmon and Knab (2018) ascribe to the business model. Read through our proposed framework, the bridge in our empirical case is built asymmetrically; value creation, delivery, and capture are reworked through operational and, to some degree, tactical work, whereas the value proposition, the element Teece (2010) treats as foundational and most closely linked to strategy, remains largely untouched. Bridging is not an all-or-nothing property of a business model but a partial, sequenced accomplishment, and the elements left unaddressed indicate how far a niche innovation has, in fact, traveled toward the new regime.

For transition management and the multi-level perspective, the study opens the firm-level black box in the early phases of transition, as the distinction between operational, tactical, and strategic work (Kemp et al., 2007) is usually applied at the system level. By locating each challenge both in a level of work and in an element of the value logic, we show that goal orientation at the project level (Geels and Schot, 2007) does not travel down into the business models of the participating actors: the project has a vision, while few individual actors express that they have a strategy. This offers a concrete explanation for why the experimentation phase of transitions (Geels and Locatelli, 2024) can appear highly active yet leave the incumbent regime unchallenged.

For the ecosystem literature, our case highlights the dissonance among different types of ecosystems (Cobben et al., 2022; Järvi et al., 2018) as actors navigate between individual firms' goals of sustainable business innovation and the shared ambitions of knowledge creation and electrified transport. The project has a clear goal, but decomposing the members' business-model work, however, shows that this shared ambition is not translated into proposition-level work within participating organizations. Facilitating, aligning, and orchestrating envisioning work (see e.g. Reypens et al., 2021; Walrave et al., 2018) in large-scale transitions remains a question for future research.

Empirically, the study addresses the scarcity of research on battery electric vehicles in forestry transport (Iyer et al., 2025) by providing qualitatively rich evidence from 17 informants representing the full value chain of a frontrunner demonstration project. In doing so, it also speaks to transport research modeling the potential reduction in CO2 emissions through electrification (e.g. Lipiäinen et al., 2022; Nicolaides et al., 2018). Such estimates rest on the assumption of adoption, where our study sheds light on the barriers to transforming the incumbent transport system, expanding the dominant focus of technical barriers through interdisciplinary research (Sugihara et al., 2023).

For practice, our findings suggest four priorities for entering an electrification project. First, make the value proposition an explicit agenda item from the outset, as it may not surface from operational problem-solving on its own. Second, translate the project's collective ambition for goals at the level of each participating organization. Third, be aware that pilot-phase support masks the question of how costs and risks will be distributed, and that the arrangements that work during a demonstration may not survive scaling. Lastly, be inclusive when it comes to stakeholders such as drivers, as they absorb much of the operational change, and their acceptance impacts the feasibility of electrified transport.

6. Conclusion

This study set out to understand the business-model challenges facing actors in the early stages of electrifying forestry transport. Combining the business model's bridging role between niche and regime with the distinction of operational, tactical, and strategic work, we located each challenge both in an element of the business model and in the level at which actors address it. Analyzed this way, the five categories of challenges we identify – technical, economic, changes in roles and collaboration, logistics, and internal organizational – resolve into a clear pattern. Actors are actively reworking value creation, delivery, and capture, largely through operational and, in matters of collaboration, tactical work. The value proposition is only sparsely addressed. The project has an explicit collective vision, yet few of the individual organizations have a corresponding strategy. Electrification, in this case, risks being absorbed into the existing business logic rather than prompting a reconsideration of what an electrified transport business offers, and to whom. Therefore, the emission reduction expected from electrified heavy transport depends not only on the vehicles and infrastructure, but on whether the actors who must operate the technology can build viable business models around it.

We identify six research avenues that follow from this study. First, the gap between project-level ambition and actor-level strategy invites research on how collective goals translate, or fail to translate, into the business models of participating organizations, and the alignment mechanisms that might support this. Second, the sparse attention to value propositions suggests a need to examine how electrification could create new forms of value, and for whom. Third, our findings that actors reason about the transition in cost-facing terms point to sensemaking as a promising lens. Fourth, since this network has no central orchestrator, power asymmetries and their implications for network coordination, risk sharing, and strategic and envisioning work merit deeper empirical exploration to facilitate the transition. Fifth, public charging for heavy vehicles depends on utilization and is currently supported by public funding. The viability of the charging infrastructure, therefore, needs to be studied over time as the number of vehicles grows and arrangements shift. Finally, the influence of truck drivers in the BEV adoption process should be further investigated to understand their perspectives, incentives, and everyday needs.

The study has limitations. Firstly, it largely relies on the perspectives of selected informants. Though they were well informed about the electrification of forestry transport and the analysis was validated with

the project participants, their personal opinions and experiences shaped the results. Second, the challenges outlined in this article represent a snapshot in time, specifically early in the transition, and it is therefore not possible to judge their overall impact on the success of the transition at large. Third, the study focuses on a specific frontrunner project, which naturally means that the organizations involved are early adopters and that mechanisms are in place to facilitate coordination and collaboration. Therefore, organizations without this context may experience different challenges. Fourth, the framework we propose is developed and applied only to a single case, so its dimensions and the coding of challenges into them are analytical choices that need further verification, ideally over time. Lastly, the project takes place in Sweden, a nation with a significant legacy in forestry and forestry transport. Different regulatory settings, infrastructure, and surrounding organizations will likely impact the challenges for electrified forestry transport.

CRediT authorship contribution statement

Johan Arekrans: Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Gunilla Ölundh Sandström:** Writing – review & editing, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Jennie Björk:** Writing – review & editing, Supervision, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used ChatGPT 5.3, Claude, and Grammarly to improve the article's structure, check grammar, and assist with translation from Swedish to English. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. – Interview guideline

This version has been translated from Swedish by the authors.

[Introduce the background to the interview, ask for questions, inform about the use of the data and ask for consent to record].

Part 1: Understanding the Current State.

This section aims to understand the present-day operations, the company's role and value proposition at the site, and which actors and relationships are important today.

A. The interviewee
   Can you describe your role within the organization? What are your main responsibilities? How long have you been working at [company]?

B. The company's role at the site today
   Can you briefly describe [the company's] operations today? Specifically, within the site defined in TREE as a test phase?
   What are you doing within TREE and why? What part of your operations will be affected by the TREE project and electrification during the test phase?
   What factors most influence your decision to introduce electric vehicles? / If you're not directly responsible for vehicles – what factors do you think most influence the decision to adopt electric vehicles in the forestry industry?

C. Current value proposition
   What is the logic behind the offerings you provide? Which factors influence your costs and revenues?
   How are you affected by other actors in delivering your value proposition? What is needed for you to deliver your offering?
   Are there factors in their value logics that affect how your value logic is structured? (e.g. governed route choices? Tonne-kilometer metrics?)
   What opportunities do you see for new business connected to electrification?
   What needs or challenges do you foresee related to business model issues and your site? What risks exist?

D. Current actors and relationships
   Thinking about your current situation—who are the actors you mainly work with or rely on to make [the operations affected by TREE] work?
   What do these collaborations look like? What is the nature of the dialogue? Who do you have discussions with?
   How is electrification driven at your site specifically? Which actors are leading? How and why?
   What needs or challenges do you see in managing your current relationships? What risks exist?

[Researchers show and asks the interviewee to reflect on the site mapping from the first workshop].

What do you think about the mapping? Does it reflect reality?
Would you remove anything? Add anything?
Have any new actors emerged?

Part 2: Understanding the future.

This section aims to understand how electrification may affect future roles, value offerings, and relationships at the site.

E. The company's role at the site in the future
Looking 10 years ahead—after the TREE project—how do you think your operations will look? What impact will electrification have had?
What will the fleet and routes look like, in your view?
What do you think will be the next step in terms of business models, following the pilot you are part of in TREE, i.e., in the medium term?
How do you see the coexistence of diesel and electric vehicles working?
What needs or challenges do you foresee in taking on your future role? What risks do you anticipate?

F. Future value proposition
What value do you create in the electrification of the forestry industry?
How will the logic of your offering change post-electrification? Which factors will influence costs and revenues?
How will you be affected by other actors in delivering your future value proposition? What will be required to deliver it?
What needs to be resolved for this to be viable on your end? (Uncertainties, unknown factors, risks, challenges)

G. Future actors and relationships
Continuing with the future perspective: which actors do you think will have been key players in making your electrification successful?

What role will new actors and relationships play, in your opinion? Those that are not currently involved?
What needs or challenges do you foresee in managing the relationships you will need? What risks do you anticipate?

H. On the decision to introduce electric vehicles – additional question
- Which factors have the greatest impact on your decision to adopt electric vehicles?

I. Closing

Is there anything you would like to add on the subject?
Are there others within your company that you think would be valuable for us to talk to?
Are there others at the site who you think are important for us to interview?
May we contact you again if we have any follow-up questions from today's interview?

Data availability

The datasets generated and analyzed during the current study are not publicly available due to the nature of the qualitative research. The data consist of interview transcripts containing potentially identifiable information. Sharing this material would compromise the anonymity and confidentiality of the participants and would not be consistent with the terms of informed consent provided. To protect the privacy and trust of the informants, and in accordance with ethical research guidelines, the data cannot be made publicly accessible.

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